We can only confirm this result for the last 70 years (and not for the last 200 years). However, when considering the studies by Price (1965) and by van Raan (2000) and our study, it should be noted that scientific growth since the beginning of modern science has not been measured directly with the publication volume, but indirectly with the number of cited references. Unfortunately, there is currently no literature database containing every publication since the beginning of modern science to today and which can be used for statistical analysis.

Our results support estimates such as those made in a comment piece by Frazzetto (2004) about the far-reaching changes that have taken place in science over the last century:

> "Without doubt, science has put on a new face in the past century. It has come to occupy a central role in society and now enjoys a privileged position among the knowledge-producing disciplines. As a consequence, the resources devoted to science have increased hugely. There has also been, in general, a visible shift in the way in which science is organized and in how it produces knowledge. Science has become 'big', global and complex … and the emphasis has shifted from the individual scientist to collaborative work. Similarly, scientific knowledge production has become more costly, and forms of funding have subsequently changed, with industry and venture capital now providing the bulk of financial resources" (Frazzetto, 2004, p. 19).

An end to this development cannot be foreseen (with new highly-productive players such as China and India moving swiftly into the science community).

Finally, we would like to mention three limitations of our study:

The first limitation concerns the use of publications to measure scientific growth. There are advantages and disadvantages to using this data, as described by Tabah (1999): "Although counting publications is simple and relatively straightforward, interpretation of the data can create difficulties that have in the past led to severe criticisms of bibliometric methodology … The main problems concern the least publishable unit (LPU), disciplinary variance, variance in quality of work, and variance in journal quality" (p. 264). Publications are not singular entities that have static forms. Publication practices differ, not only between disciplines, but also within disciplines. Publication numbers and both publishing styles as well as citing styles have developed through history (de Bellis, 2009). Add to this the explosion in publishing numbers linked to external factors in the past decades, related to the “Audit Society” and “New Public Management”, with “salami sliced publishing”, and the notion of “least publishable unit” (Bornmann & Daniel, 2007). Not to mention that at least during the first 100 years of publishing in *Philosophical Transactions*, it was not the scientists that published their work, but the editor, that received letters of talks read to the community that were published. There is of course the option of using other data than publications to measure scientific growth such as the number of scientists. However, as a rule this other data is not more suitable than bibliometric data as it has its own limitations (for example, there is no database that can provide reliable information about the number of scientists practising modern science since its beginnings to this day).

The second limitation refers to the view of “growth” as an “increase in numbers” (publications, cited references, or scientists). This study views science through an internal perspective, where all differences are expected to be the result of internal aspects of a static research practice in “modern science”, as opposed to a mixture of internal (analytic) practices and external (sociological, historical, psychological) practices that continuously have altered the ways of viewing science in the same four centuries that this study focuses on. Furthermore, it is not clear whether an “increase in numbers” is directly related to an “increase of actionable knowledge”, for example for reducing needs, extending our knowledge about nature in some lasting way or some other “higher purposes” (Bornmann, 2012, 2013).

The third limitation concerns the database used here: WoS. According to van Raan (2000) “characteristics of the instruments (in this case: the database) will also be reflected in the measurements. Even worse, characteristics of databases may change in the course of time”

(p. 348). The WoS (or the SCI) has been in operation for several decades and has changed over the years. For example, the coverage of journals has been extended. The growth of science is not the only explanation for an increasing number of publications in the data, as Michels and Schmoch (2012) note: "By subdividing the journals into different categories, it was possible to distinguish which increases are related to a growth of science and which to the database provider's policy of achieving a broader coverage of journals. The number of articles in the observation period [2000-2008] grew by 34 percent in total, a remarkable increase for this short period. If the number of articles in the categories 'start' and 'new', which represent the growth of science, are added together, the growth rate in this period is 17 percent" (p. 841). As this study is mainly based on an analysis of cited references, we are assuming that the changes of the database characteristics do not exert a major influence on the results. As we have shown, even substantial changes to the data, such as limiting it to one year or one field in the citing papers, does not have a significant influence on the results arrived at with the cited references.

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